INCH-POUND

MIL-W-19044E(SH)
AMENDMENT 1
18 July 1994

MILITARY SPECIFICATION

WASHER-EXTRACTOR, LAUNDRY, 35 TO 200 POUND CAPACITY, MODIFIED COMMERCIAL (SHIPBOARD)

This amendment forms a part of MIL-W-19044E(SH), dated 18 September 1993 and is approved for use by all Departments and Agencies of the Department of Defense.

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1.2: Delete the following Classification:

"Class 1 - Electro-mechanical automatic formula control unit."

1.2: Add the following Classification:

"Class 3 - Programmable logic control automatic formula control unit"

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2.1.2, lines 7 and 8: Delete “Navy Resale and Service Support Office, Fort Wadsworth, Staten Island, NY 10305” and substitute “Navy Exchange Service Command, Fleet Industrial Supply Center, Norfolk, VA 23511.”

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2.2: Add the following Non-Government publication:

"UNDERWRITERS LABORATORY, INC.
UL 508 - UL Standard for Safety Industrial Control Equipment (Fifteenth Edition)

(Applications for copies should be addressed to UL Publications Stock, 333 Pfingsten Road, Northbrook, Illinois 60062.)"

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3.3.1.1, lines 5 and 6: Delete "electro-mechanical type on class 1 washer-extractors and solid state, microprocessor type on class 2 washer-extractors" and substitute "solid state, microprocessor type on class 2 washer-extractors and programmable logic control type on class 3 washer-extractors."

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3.4.8.1, lines 1 through 3: Delete "of the replaceable plastic chart type on class 1 washer-extractors and a microprocessor controller on class 2 washer-extractors" and substitute "a microprocessor controller on class 2 washer-extractor and a programmable logic controller on class 3 washer-extractors".

3.4.8.1.2, lines 1 through 6: Delete.

3.4.8.1.3: lines 12 and 13: Delete "Indicators shall provide analog or digital display of the set point bath temperature" and substitute "Indicators shall provide digital display of the actual bath temperatures during the wash and rinse cycles and the set points of the bath temperatures of the formula in progress."

3.4.8.1.4: Delete and substitute:

"3.4.8.1.4 Step and Time Display. Class 2 and class 3 washer-extractors shall have a digital program formula timer and a program formula step display indicating time left within the formula and formula step in operation. The timer shall be descending in one minute increments."

3.4.8.1.11, Make the following changes:

line 1: Delete "may" and substitute "shall".
line 2: Delete "remotely or". Delete "When the" and substitute "The".
lines 2 and 3: Delete "is directly mounted it".
lines 6 and 7: Delete "The controller shall provide a minimum of 8 feet of cable length for remote mounting."

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Add as new paragraph 3.4.8.1.13:

"3.4.8.1.13 Programmable logic controller. The programmable logic control automatic formula control unit shall be preprogrammed with three standard Navy wash formulas published in Navy Exchange Service Command Ship's Store Bulletin and one short cycle test formula which demonstrates each phase of operation. Each formula shall be energized by depressing a single, individual push-button located on the front of the control panel for each of the four formulas. A means shall be provided to lock-out the test formula to prevent its use except during testing and maintenance operation. Program memory shall be stored in EEPROM memory. The EEPROM ROM cassette shall be mounted to the programmable logic controller and shall be readily replaceable to accommodate program changes. Programmable logic controller shall meet the applicable requirements of UL 508."

Add as new paragraph 3.4.8.1.13.1:

"3.4.8.1.13.1 Displays, buttons and switches on controller. Cycle indicator lights, power on, emergency stop and low air lights with lense caps, and temperature, program time and program step indicators shall be displayed on the front of the controller. Emergency stop and jog push-buttons and informational label plate specifying formula classification and operational instructions shall be installed on the front of the control panel."

Add as new paragraph 3.4.8.1.13.2.:

"3.4.8.1.13.2 Manual operation. An integrated female pin connector for manual over-ride operation shall be located in an accessible area on the back of the control panel electrical box enclosure. The manual operation shall be independent of the programmable logic controller and shall individually control hot and cold water valves, wash, high extract, drain and brake. The phases of the manual operation shall be controlled by a remote, hand held controller with a male pin connector and individual switches corresponding to each phase of manual operation. The female pin connector shall be threaded and provided with a protective cap."

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TABLE V, lines 4 and 5: Delete "pin connector to microprocessor not as specified (see 3.4.8.1.2)." and substitute "Programmable logic controller not as specified (see 3.4.8.1.13 and 3.4.8.1.13.1)."

Preparing activity
NAVY-SH
(Project 3510-N340)

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